# Intro to LaTeX

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# Common Document File Formats

**txt:** is just a sequence of bytes.

Advantages: simple, lightweight.

Disadvantages: No formulas, images or general typesetting.

## An Example

```
Suppose a particle moves along a straight line with a constant acceleration $a$. Its position at time $t$ can be described by the equation
[
x(t) = x_0 + v_0t + \frac{1}{2}at^2,
]
where $x_0$ is the initial position and $v_0$ is the initial velocity. If the particle starts from rest at $x_0=0$ and accelerates at $a=2,\mathrm{m/s^2}$, its position after $t=3,\mathrm{s}$ is
[
x(3) = \frac{1}{2}(2)(3^2) = 9,\mathrm{m}.
]
This simple example shows how mathematical models can describe the motion of a physical system.
```

# Common Document File Formats

**docx:** stores the content, structure, and formatting of an editable document.

Advantages: easy to edit, support rich typesetting.

Disadvantages: unstable and incovenient fomulas, low portability.

## On My Computer

Suppose a particle moves along a straight line with a constant acceleration $a$. Its position at time $t$ can be described by the equation

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2,$$

where $x_0$ is the initial position and $v_0$ is the initial velocity. If the particle starts from rest at $x_0 = 0$ and accelerates at $a = 2\,\mathrm{m/s^2}$, its position after $t = 3\,\mathrm{s}$ is

$$x(3) = \frac{1}{2}\cdot 2 \cdot 3^2 = 9\,\mathrm{m}.$$

This simple example shows how mathematical models can describe the motion of a physical system.

## On My Phone

Suppose a particle moves along a straight line with a constant acceleration . Its position at time can be described by the equation

where is the initial position and is the initial velocity. If the particle starts from rest at and accelerates at , its position after is

This simple example shows how mathematical models can describe the motion of a physical system.

# PDF File Format

PDF (Portable Document Format) is a file format designed to preserve the "exact typesetting" of a document.

## On My Computer

Suppose a particle moves along a straight line with a constant acceleration $a$. Its position at time $t$ can be described by the equation

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2,$$

where $x_0$ is the initial position and $v_0$ is the initial velocity. If the particle starts from rest at $x_0 = 0$ and accelerates at $a = 2\text{m/s}^2$, its position after $t = 3\text{s}$ is

$$x(3) = \frac{1}{2} \cdot 2 \cdot 3^2 = 9\text{m}.$$

This simple example shows how mathematical models can describe the motion of a physical system.

## On My Phone

Suppose a particle moves along a straight line with a constant acceleration $a$. Its position at time $t$ can be described by the equation

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2,$$

where $x_0$ is the initial position and $v_0$ is the initial velocity. If the particle starts from rest at $x_0 = 0$ and accelerates at $a = 2\text{m/s}^2$, its position after $t = 3\text{s}$ is

$$x(3) = \frac{1}{2} \cdot 2 \cdot 3^2 = 9\text{m}.$$

This simple example shows how mathematical models can describe the motion of a physical system.

# What is LaTeX?

LaTeX is a document preparation system that uses source code to produce professionally typeset PDF files, good at:

- Mathematical formulas;
- Citation managements;

Every student is required to submit your homeworks using LaTeX :)

# How to Use LaTeX?

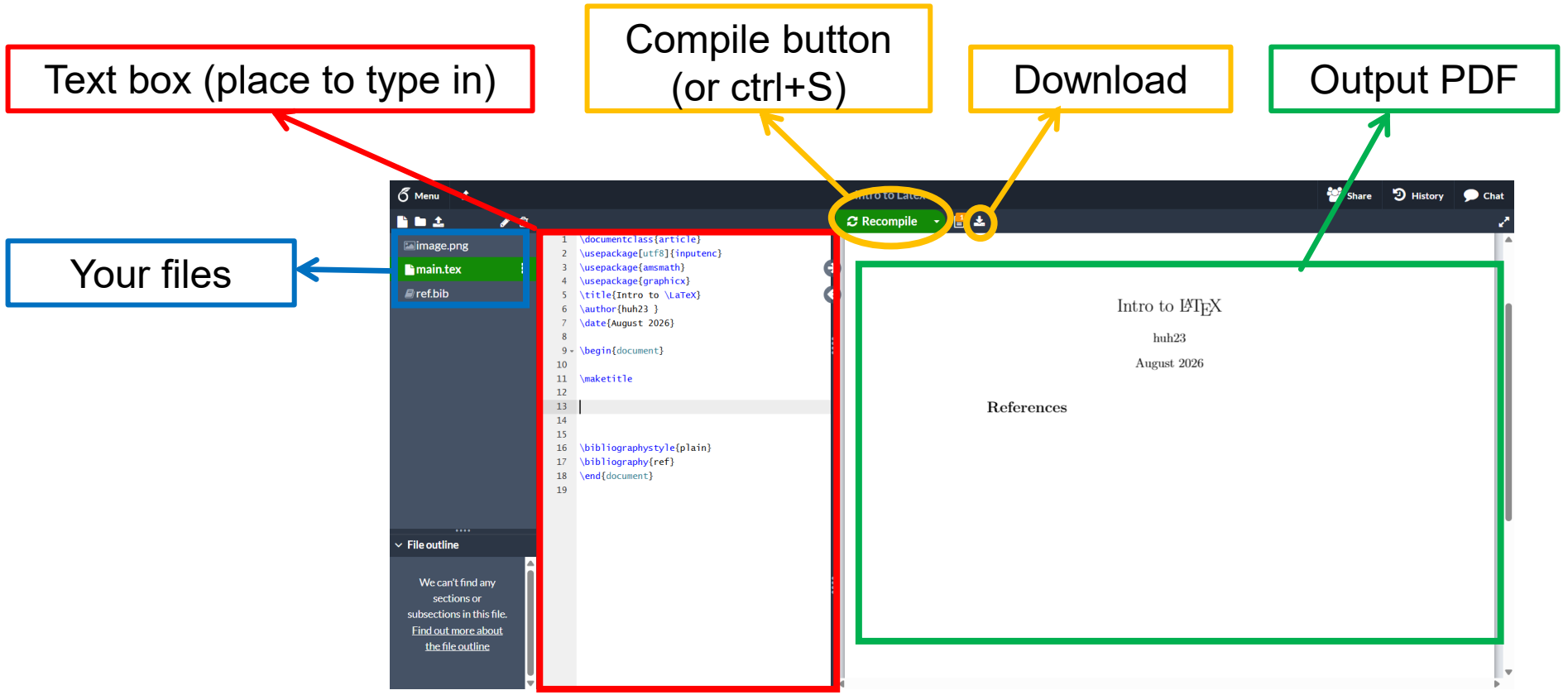
In general, two ways:

1. Install a LaTeX compiler, write and compile locally (not recommended)

2. Overleaf --- Online LaTeX editor, no installation required (https://overleaf.tsinghua.edu.cn) **(recommended)**

# LaTeX --- A Simplest Project

Let’s open a blank project. You can easily find the correspondence.



# Line Breaks and Spaces

- A line break = no line break.
- Two or more line breaks = one new paragraph.
- \\ to break a line without indentation.
- Multiple spaces = one space.
- Spacing in LaTeX includes \,, \:, \;, \ , \quad, \qquad, \enspace, \hfill, and \hspace{3cm}.

# Sections

- \section{} is numbered.
- \section*{} is unnumbered.

# Math Formula

- $xyz$: inline math environment.
- Superscript: $x^2$
- Subscript: $H_2O$
- Free spacing: $x    +     y = z$
- $$P\ne NP$$ denotes independent math environment, but now we recommend to use \[P\ne NP\].
- Other examples: $\frac{1}{x}, \sqrt{x}, \sqrt[3]{x}, \le, \ge, \ne, \approx, \pm, \times, \cdot, \cdots$

# Math Formula

- Greek alphabet: $\alpha, \beta, \gamma, \delta, \theta,  \Gamma, \Delta, \Theta, \cdots$
- Adaptive brackets: \[ \left( \frac{x}{y} \right) \] \[ \left[ \frac{x}{y} \right\} \]
- Matrix:

```
48  \[
49  \left[
50  \begin{matrix}
51  1 & 2 \\
52  3 & 4 \\
53  \end{matrix}
54  \right]
55  \]
```

# Math Formula

- Multiple-line formula:

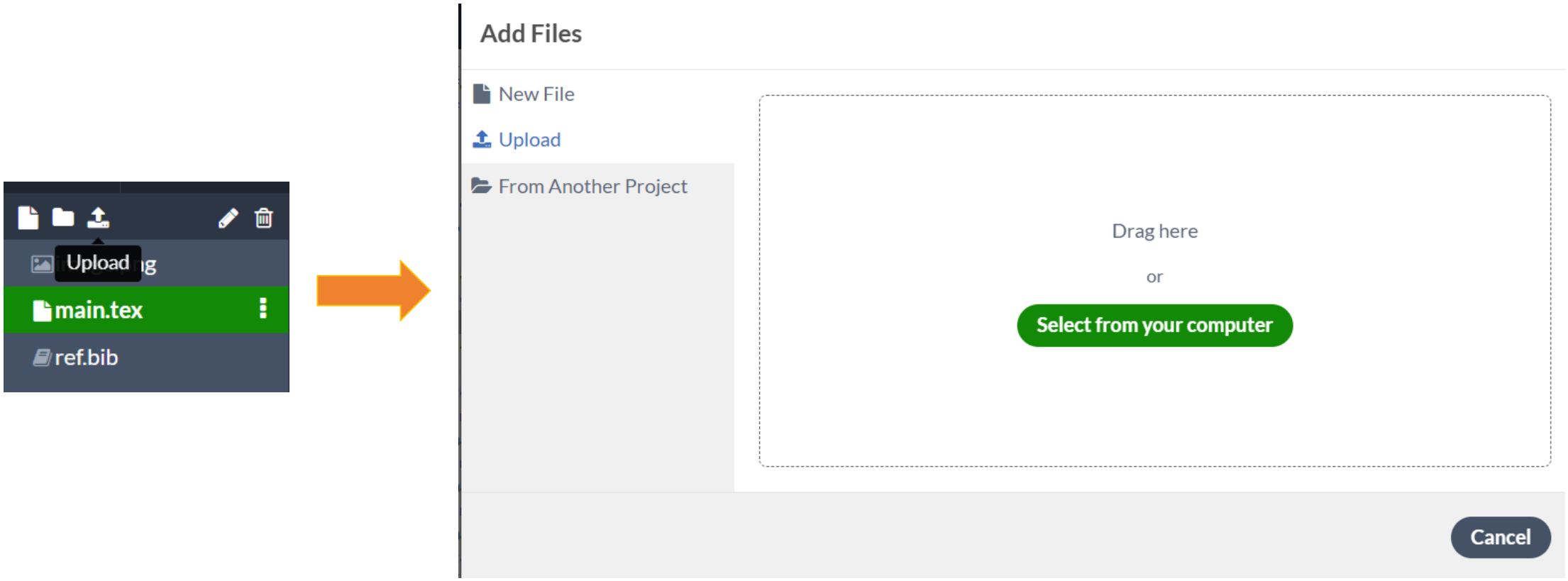
```
\begin{align}
    E &= mc^2 \\
    e^{\pi i} &= -1
\end{align}
```

- If you want to recall a formula:

```
\begin{align}
    E &= mc^2 \label{eq:eins} \\
    e^{\pi i} &= -1 \label{eq:euler}
\end{align}
I prefer Eq. \eqref{eq:eins} to Eq.
\eqref{eq:euler}.
```

# Figures

- Upload a figure:



# Figures

- Place a figure in a simple way: \includegraphics[width=0.33\linewidth]{image.png}
- Place a figure in a serious way:

```
77  \begin{figure}[htbp]
78      \centering
79      \includegraphics[width=0.33\linewidth]{
        image.png}%
80      \includegraphics[width=0.33\linewidth]{
        image.png}%
81      \includegraphics[width=0.33\linewidth]{
        image.png}
82      \caption{An example figure.}
83  \end{figure}
```

# Figures

- [htbp] specifies the preferred placement of a floating figure or table:
  - h: here
  - t: top of the page
  - b: bottom of the page
  - p: a separate page for floats

# Tables

- Directly type the elements.

```
\begin{tabular}{ccc}
A & B & C \\
1 & 2 & 3
\end{tabular}
```

- Add vertical lines:

```
\begin{tabular}{c|c|c}
A & B & C \\
1 & 2 & 3
\end{tabular}
```

- Horizontal lines:

```
\begin{tabular}{c|c|c}
\hline
A & B & C \\
\hline
1 & 2 & 3 \\
\hline
\end{tabular}
```

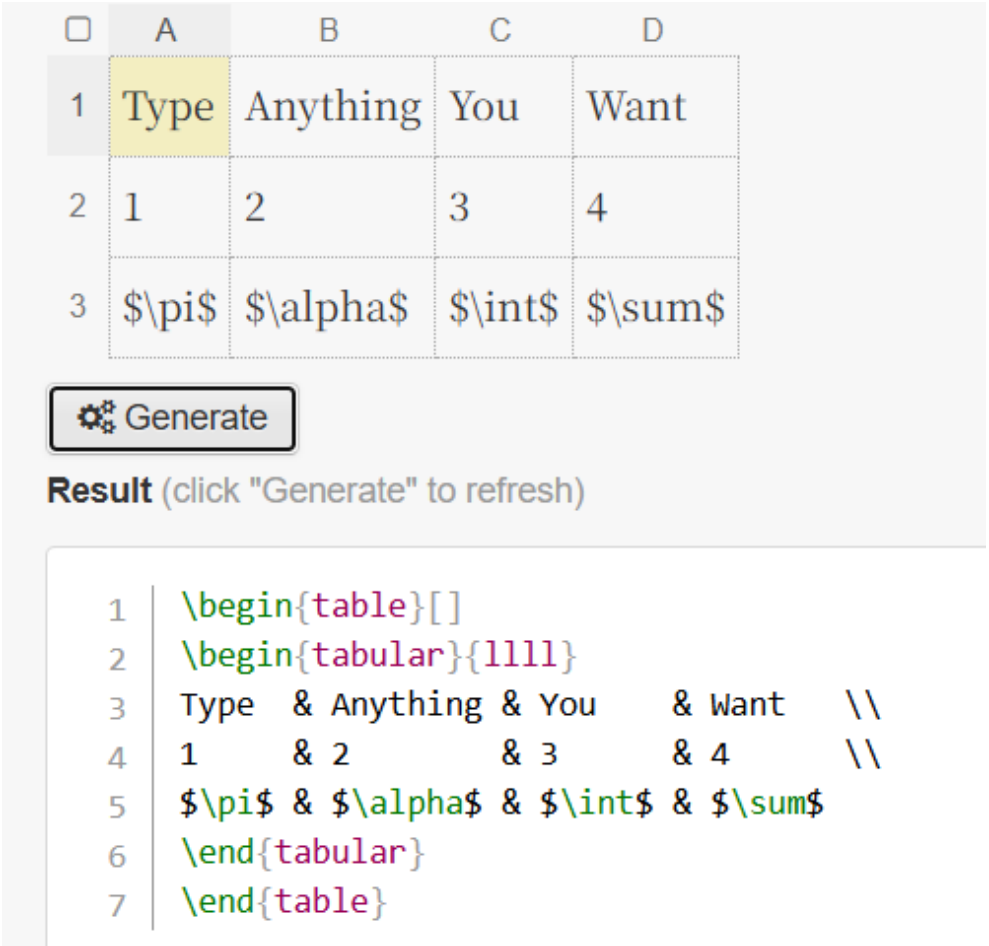
# Tables

- A more formal environment:

```
\begin{table}[htbp]
    \centering
    \begin{tabular}{c|c|c}
        \hline
        A & B & C \\
        \hline
        1 & 2 & 3 \\
        \hline
    \end{tabular}
    \caption{Example table}
    \label{tab:example}
\end{table}
Table \ref{tab:example} is an example
table.
```

# Tables

- Alternative way: use https://www.tablesgenerator.com/.



# References

- 1. Search scholar.google.com.
- 2. Select a paper.

(1)



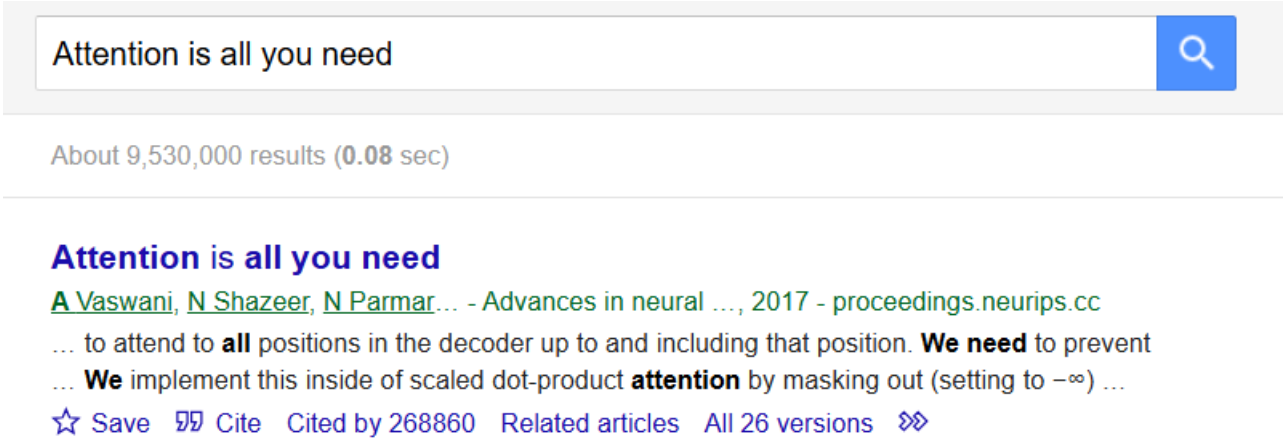

(2)



# References

- 3. Click “cite”.
- 4. Click “BibTex”.

(3)




(4)



# References

- 5. Copy it into your .bib file.

(5)

```
@article{vaswani2017attention,
  title={Attention is all you need},
  author={Vaswani, Ashish and Shazeer, Noam and Parmar, Niki and Uszkoreit, Jakob and Jones, Llion and Gomez, Aidan N and Kaiser, {\L}ukasz and Polosukhin, Illia},
  journal={Advances in neural information processing systems},
  volume={30},
  year={2017}
}
```

# References

- References are usually stored in a .bib file.
- After that, you can cite an academic paper: The Transformer architecture was introduced \cite{vaswani2017attention}.
- Don’t forget

```
135  \bibliographystyle{plain}
136  \bibliography{ref}
```

# Supplementary Information

To visit the project I've shown in the class:

https://overleaf.tsinghua.edu.cn/read/dccfspkbgqpp

Latex tutorial:

https://latex-tutorial.com/